Here, the total number of students to be selected is 10 and if only two girls are interested to participate in the competition, then how many boys should be selected? Obviously, if y = 2 then x + 2 = 10. The value x = 8 satisfies the equation. So, the number of boys to be selected is 8.

Follow the steps to obtain the value.

Step – 1: Study the problem. Fix the variable and write the algebraic expression.

Step – 2: Replace each variable by the given numerical value to obtain an arithmetical expression.

Step – 3: Simplify the arithmetical expression by **BIDMAS** method.

Step – 4: The value so obtained is the required value of the expression.

Try this

Try to find the value of the following expressions, if p = 5 and q = 6.

i) $p + q$ ii) $q - p$ iii) $2p + 3q$ iv) $pq - p - q$ v) $5pq - 1$

Example 3.3

If x = 3, y = 2 find the value of (i) $4x + 7y$ (ii) $3x + 2y - 5$ (iii) $x - y$

Solution

i) $4x + 7y = 4\,(3) + 7\,(2) = 12 + 14 = 26$

ii) $3x + 2y - 5 = 3\,(3) + 2\,(2) - 5 = 9 + 4 - 5 = 8$

iii) $x - y = 3 - 2 = 1$

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Example 3.4

Find the value of (i) $3m + 2n$ (ii) $2m - n$ (iii) $mn - 1$, given that m = 2, n = -1.

Solution

i) $3m + 2n = 3(2) + 2(-1) = 6 - 2 = 4$

ii) $2m - n = 2(2) - (-1) = 4 + 1 = 5$

iii) $mn - 1 = (2)(-1) - 1 = -2 - 1 = -3$

Exercise 3.1

1. Fill in the blanks:

(i) The variable in the expression $16x - 7$ is _____.

(ii) The constant term of the expression $2y - 6$ is _____.

(iii) In the expression $25m + 14n$, the type of the terms are _____ terms.

(iv) The number of terms in the expression $3ab + 4c - 9$ is _____.

(v) The numerical co-efficient of the term $-xy$ is _____.

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